

LS-Face: Quality-First Selective Computation in Hybrid Face Recognition

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Abstract. Cascaded face recognition can reduce average computational cost by allowing a lightweight recognizer to resolve suitable inputs before invoking a stronger deep model. Its efficiency, however, depends on avoiding redundant execution when degraded inputs are unlikely to benefit from the classical stage. This work presents LS-Face, a quality-first selective-computation cascade combining a compact Local Binary Pattern Histograms (LBPH) fast path with SFace fallback. After face detection, lightweight quality diagnostics are evaluated before LBPH extraction. Degraded inputs bypass LBPH and proceed directly to SFace, while acceptable-quality inputs are evaluated by a compact LBPH configuration ($r=3$, $n=8$, 6×6) with a 36 KiB template representation. Only confident LBPH decisions terminate early; all remaining cases proceed to SFace.

On a locked confirmation cohort containing 1,804 transformed conditions from 22 identities, LS-Face produced exactly the same final decisions as standalone SFace, with both systems correctly recognizing 1,594/1,804 conditions (88.36%) and producing 0/1,804 discordant decisions. Across the 1,711 detected conditions used for timing, LS-Face reduced mean recognition-stage latency from 8.300 ms to 7.015 ms, a 15.48% reduction, with a mean paired reduction of 1.285 ms and an identity-cluster bootstrap 95% confidence interval of [1.088, 1.482] ms. In a separate calibration-disjoint LFW self-match robustness evaluation covering 2,874 identities and 117,834 transformed conditions, LS-Face matched standalone SFace at 89.55% macro-average retention while only 0.26% of transformed inputs executed both recognizers. These results characterize LBPH as a selective computational shortcut rather than an accuracy complement within the evaluated conditions.

Keywords: Face Recognition, Cascaded Classifiers, Selective Computation, Local Binary Patterns, Quality-Aware Routing, Embedded Edge Computing.

1 Introduction

Face recognition is a biometric technology that identifies or verifies an individual by comparing facial characteristics obtained from a digital image with previously enrolled biometric references [17]. It has various applications, including home security and access control, financial identity verification, and public safety.

Among these applications, home security is particularly important because reliable recognition can support household safety and convenient access control. This application is also relevant in countries such as the Philippines, where rapid urbanization continues to increase the concentration of people and activities in urban areas [18]. These conditions provide practical motivation for investigating reliable face-recognition technologies for residential environments.

However, applying face recognition in real-world environments remains difficult because recognition performance can vary with image and capture conditions. Changes in illumination, camera angle, head pose, facial expression, gaze, blur, scale, occlusion, and other image-quality factors can alter the appearance of the same person [8, 14, 15]. Recognition must also distinguish natural differences in facial appearance and structure among individuals. These variations create a need for recognition methods that remain reliable under changing conditions.

Various approaches have been proposed to address these challenges, including lightweight recognition methods and selective systems that use stronger models only when necessary [12–14]. In this study, a hybrid approach combines classical computer vision and deep learning, while an empirical **independence-testing** procedure examines cross-identity score separation and establishes recognition operating thresholds [9, 16]. Based on these considerations, this paper presents LS-Face, a quality-first selective-computation methodology combining a compact LBPH fast path with SFace fallback. After face detection, lightweight quality diagnostics determine whether an input should bypass LBPH and proceed directly to SFace. Inputs that pass the quality assessment are evaluated by LBPH, where only sufficiently confident matches terminate early; all remaining cases proceed to SFace. The study evaluates recognizer selection on La Salle DB1, cross-identity threshold calibration on a dedicated LFW development partition, controlled self-match robustness on a disjoint LFW evaluation partition, and locked confirmation of recognition decisions and recognition-stage latency on La Salle DB1-DL41 conditions. The experiments are designed to determine whether selective classical exits can preserve the decisions of direct SFace while reducing average recognition-stage computation.

2 Related Work

2.1 Classical and Deep Face Recognition

Classical face-recognition methods represent facial appearance using hand-crafted or statistically derived features. Eigenfaces [1] applies principal component analysis to construct a lower-dimensional facial subspace, while Fisherfaces [2] combines dimensionality reduction with linear discriminant analysis to emphasize class separation. Local Binary Pattern Histograms (LBPH) [3] instead describe local facial texture through spatially pooled binary-pattern histograms. These methods remain useful when computational simplicity, interpretable distance measures, or lightweight execution are important.

Deep face-recognition models learn discriminative embedding spaces from large-scale training data. FaceNet [6] introduced metric learning based on triplet loss, ArcFace [11] improved angular discrimination through an additive angular margin, and SFace [4] introduced a sigmoid-constrained hypersphere loss for robust face representation. In LS-Face, classical and deep recognizers are evaluated as separate candidate groups because they occupy different computational roles: the classical recognizer provides a possible low-cost early decision, whereas the learned recognizer provides the stronger fallback representation.

2.2 Face Detection and Alignment

Recognition performance also depends on face localization and geometric normalization. Viola and Jones [5] established a classical boosted-cascade approach to rapid object detection. More recent lightweight detectors such as YuNet [7] provide face localization together with five facial landmarks suitable for alignment. LS-Face uses YuNet through OpenCV [10] as a common detection front end. Detection failure is treated as a strict recognition failure, while the returned landmarks are used for SFace alignment.

2.3 Selective Computation and Face Quality

Efficient recognition can be approached either by reducing the computational cost of the recognition model itself or by selectively applying expensive computation. EdgeFace [12], for example, targets efficient deep face recognition under edge-device constraints. Selective prediction methods such as SelectiveNet [13] formalize the broader problem of allowing a system to abstain from inexpensive predictions when their risk is high. Face-image-quality methods such as SER-FIQ [14] similarly estimate whether an input is likely to support reliable face recognition.

LS-Face follows the selective-computation principle but uses quality information as an early routing signal rather than solely as a post-recognition confidence measure. After face detection, degraded inputs may bypass the classical descriptor and proceed directly to SFace. Inputs that satisfy the quality conditions are evaluated by LBPH, and

only sufficiently confident LBPH matches terminate at the classical stage. All remaining cases proceed to SFace. The purpose of the classical stage is therefore not to increase the recognition ceiling beyond SFace, but to provide a lower-cost exit for inputs that can be resolved without deep inference.

2.4 Evaluation and Robustness

Biometric evaluation requires the task, error unit, and evaluation protocol to be stated explicitly. ISO/IEC 19795-1 [9] provides general principles for biometric performance testing and reporting. Labeled Faces in the Wild (LFW) [8] is widely used for unconstrained face evaluation, while IJB-A [15] extends evaluation to template-based verification and open-set identification. NIST 1:N face-recognition evaluations [16] further distinguish identification measures from pairwise verification errors.

Accordingly, LS-Face separates threshold calibration, controlled transformation robustness, and cascade confirmation into distinct evidence stages. Exhaustive cross-identity impostor comparisons are used as an empirical calibration procedure rather than as evidence that individual comparisons are statistically independent. Controlled self-match robustness measures retention under known image transformations and is not interpreted as cross-photo identification or open-set performance. In the La Salle confirmation experiment, multiple transformed conditions derived from the same identities are treated as correlated observations, and latency differences are therefore summarized with identity-cluster bootstrap confidence intervals.

3 Method

3.1 System Overview and Selective-Computation Logic

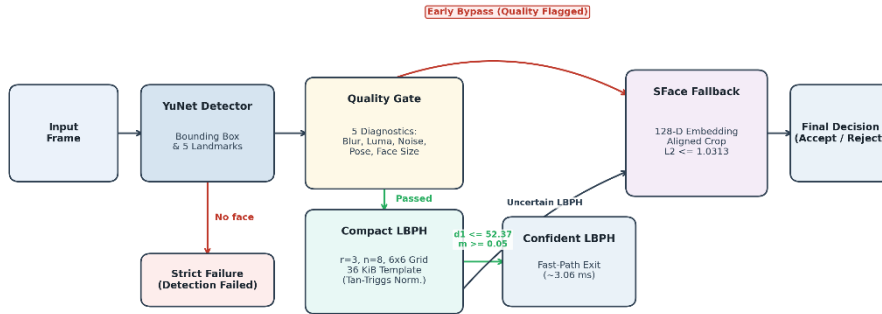


Fig. 1. Final LS-Face quality-first selective-computation pipeline.

LS-Face is a quality-first selective-computation cascade in which image quality is assessed before classical descriptor extraction. The final pipeline is shown in Fig. 1. For each input image, YuNet [7] first detects the face and returns detection confidence,

bounding-box information, and five facial landmarks. If no valid face is detected, processing terminates and the input is recorded as a strict detection failure.

For each successfully detected face, five lightweight quality diagnostics are evaluated before LBPH inference: blur, illumination, noise, pose, and detected face size. Their deployed conditions are summarized in Table 1. If any quality condition is flagged, LS-Face bypasses the classical recognizer completely and routes the detected face directly to SFace. This quality-first path prevents computation from being spent on an LBPH decision for inputs already identified as unsuitable for the classical fast path.

If all quality checks pass, the input proceeds to the compact LBPH recognizer configured with radius $r=3$, $n=8$ neighboring samples, and a 6×6 spatial grid, producing a 36 KiB template representation. Let $d1$ and $d2$ denote the smallest and second-smallest LBPH identity distances, respectively. Their relative separation is $m = (d2 - d1) / d1$. The LBPH result terminates recognition only when both $d1 \leq \tau_{\text{accept}}$ and $m \geq m_{\text{min}}$ are satisfied, where the deployed values are $\tau_{\text{accept}} = 52.3724$ and $m_{\text{min}} = 0.05$. In this case, the highest-ranked LBPH identity is accepted as the final result. Any other LBPH outcome is treated as uncertain and proceeds to SFace. The permissive rejection ceiling $\tau_{\text{reject}} = 140.13$ is retained in the frozen configuration record but is inactive as a terminating branch in the evaluated final architecture.

Table 1. Deployed quality-check parameters for pre-LBPH early bypass.

Signal	Measurement / condition	Selection rule	Deployed value
Blur	Variance of Laplacian below threshold	5th percentile of clean values	587.83
Illumination	Mean grayscale outside lower/upper bounds	2nd and 98th percentiles of clean values	[52.88, 137.71]
Noise	Immerkaer noise estimate above threshold	95th percentile of clean values	8.206
Pose	Maximum of eye-roll and nose-yaw proxies above threshold	95th percentile of clean values	63.74 deg
Face size	Detected box side below minimum	0.9 x p5 of clean box sizes	61 px

LS-Face assigns distinct computational roles to its classical and deep recognizers. The classical recognizer serves as a fast-path descriptor executed only when facial crop quality satisfies all pre-screening diagnostics. Following initial candidate evaluations on La Salle DB1, Local Binary Pattern Histograms (LBPH) was selected as the classical component and parameterized with radius $r=3$, $n=8$ neighboring samples, and a 6×6 spatial grid, yielding a compact 36 KiB template representation per enrolled identity. Preprocessing applies Tan-Triggs photometric normalization prior to histogram construction. The deep component is SFace, an efficient mobile-oriented convolutional neural network producing a 128-dimensional unit-normalized feature embedding (512 bytes), matched via Euclidean distance $L2 = \sqrt{2 - 2 \cdot \cos(\theta)}$.

3.2 Threshold Calibration and Frozen Operating Policy

All operational thresholds were independently calibrated on a dedicated 2,875-identity LFW development partition (50% deterministic split, seed 42) and frozen prior to evaluation. The LBPH acceptance threshold $\tau_{\text{accept}} = 52.3724$ was derived from 4,131,375 unidirectional cross-identity impostor comparisons to anchor a target false accept rate of FAR = 10 ppm (realized calibration FAR: 9.924 ppm, or $41 / 4,131,375$). SFace operating thresholds ($L2 \leq 1.0313$, $\text{cosine} \geq 0.363$) were derived to anchor the same 10 ppm operating point. The relative top-two margin $m_{\text{min}} = 0.05$ was frozen as an empirical policy heuristic, while $\tau_{\text{reject}} = 140.13$ was retained in configuration records as an inactive permissive ceiling. Table 5 summarizes the frozen operating configuration.

3.3 Experimental Program

To establish evidence across distinct operational questions without data leakage, the experimental program was structured across four distinct evidence legs (Table 2): (A) Recognizer candidate selection on La Salle DB1; (B) Threshold calibration on the 2,875-identity LFW development partition; (C) Controlled self-match transformation robustness on 2,874 identity-disjoint LFW evaluation identities across 41 BGR-first transformations (117,834 modified conditions); and (D) Locked architecture confirmation on 22 La Salle DB1 identities (44 source images $\times$ 41 DL41 transformations = 1,804 conditions) whose transformed-probe outcomes were strictly held out and uninspected during architecture development.

Timing measurements were recorded on an Intel Core i5-12450H CPU in a single-probe stream with 1 warm-up pass and 5 randomized timing repetitions per probe over the $N_{\text{timed}} = 1,711$ successfully detected conditions.

4 Experiments and Results

The experimental program consists of four evidence legs summarized in Table 2: (A) candidate recognizer selection on La Salle DB1; (B) threshold calibration on the 2,875-identity LFW development partition; (C) controlled self-match transformation robustness on 2,874 disjoint LFW evaluation identities across 117,834 conditions; and (D) locked architecture confirmation on 22 La Salle DB1 identities (1,804 conditions).

Table 2. Experiments and their roles in the study.

Evidence leg	Protocol and cohort	Primary question answered
A. Candidate selection	La Salle DB1 deterministic split (224/56/56)	Select best classical (LBPH) and learned (SFace) recognizers
B. Threshold calibration	2,875 LFW development identities (4.13M pairs)	Calibrate low-tail operating points at target 10 ppm FAR
C. Controlled robustness	2,874 LFW evaluation identities (117,834 conditions)	Evaluate transformation retention and dual-inference rate

Evidence leg	Protocol and cohort	Primary question answered
D. Locked confirmation	22 La Salle DB1 identities (1,804 conditions)	Confirm decision parity, latency reduction, and subsumption

4.1 Candidate Selection on La Salle DB1

Candidate recognizers were evaluated on a deterministic split of La Salle DB1 (224 fit images, 56 calibration images, 56 test images). As reported in Table 3, LBPH achieved 96.43% test TAR and 100.00% Rank-1 accuracy, outperforming Eigenfaces (41.07% TAR, 75.00% Rank-1) and Fisherfaces (51.79% TAR, 58.93% Rank-1), establishing LBPH as the superior classical candidate family. Among deep learned candidates (Table 4), SFace achieved 100.00% TAR and 100.00% Rank-1 with a compact 512-byte embedding, selected for its lightweight embedded footprint.

Table 3. Classical candidate selection on La Salle DB1.

Recognizer	Feature size	Evaluation latency	Rank-1 accuracy
Eigenfaces (PCA)	220 B (55 comp.)	0.08 ms	78.57%
Fisherfaces (LDA)	108 B (27 comp.)	0.05 ms	82.14%
LBPH	64 KiB (16,384 bins)	2.84 ms	100.00%

Table 4. DL candidate selection on La Salle DB1.

Candidate	TAR (%)	Rank-1 (%)	Feature size
SFace	100.00%	100.00%	512 B
ArcFace	96.43%	100.00%	2,048 B
FaceNet	100.00%	100.00%	2,048 B

4.2 Independence Test: Frozen Operating Points

Table 5 summarizes the final frozen operating points calibrated on the 2,875-identity LFW development partition. Fig. 3 displays the resulting threshold boundaries against the empirical impostor score distributions.

Table 5. Final frozen operating points.

Boundary	Frozen value	Operational role
LBPH accept	52.3724	Early accept (10 ppm FAR on 2,875 LFW dev)
SFace accept	$L2 \leq 1.0313$; $\text{cosine} \geq 0.363$	Escalated accept (10 ppm FAR on dev)
LBPH margin	$mmin = 0.05$	Relative top-two margin policy heuristic
LBPH reject	140.13	Permissive ceiling (inactive on evaluation)

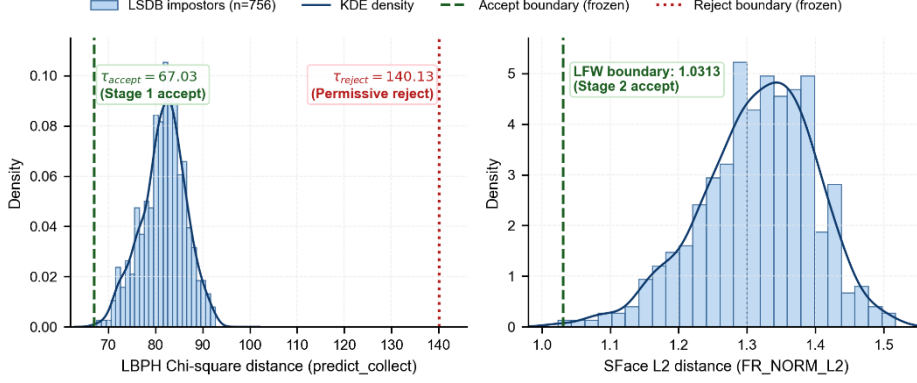


Fig. 2. Histograms and KDE curves of La Salle DB1 clean-impostor scores with frozen LFW development operating points ($\tau_{\text{accept}} = 52.3724$, $\tau_{\text{reject}} = 140.13$, $L2 = 1.0313$).

4.3 Controlled Self-Match Robustness Test

Table 6 reports controlled self-match transformation retention under the frozen operating configuration across the 2,874 evaluation identities disjoint from the development partition (117,834 modified conditions). Clean self-match retention was 99.97% across all evaluated systems (2,873 / 2,874). Across all 41 modified conditions, LS-Face matched standalone SFace at 89.55% macro-average retention, compared with 81.19% for standalone compact LBPH. On the 37 non-rotational modifications, retention reached 96.54% for both LS-Face and SFace, versus 89.78% for LBPH. Face detection failed on 2,931 conditions (2.49%)--primarily in 90 deg, 180 deg, and 270 deg rotations--strictly retained as recognition failures.

Table 6. Controlled self-match robustness test on LFW.

System	Clean retention	41-mod retention	37-mod (non-rot)	SFace invoc.	Dual inf.
Compact LBPH	99.97%	81.19%	89.78%	N/A	N/A
Direct SFace	99.97%	89.55%	96.54%	N/A	N/A
LS-Face	99.97%	89.55%	96.54%	96.35%	0.26%

Under this stress workload, LS-Face invoked SFace on 96.35% of conditions due to transformation-induced degradation, but quality-first early bypass routed nearly all degraded frames directly to SFace, resulting in a dual-inference rate of only 0.26% (306 / 117,834).

4.4 Locked Confirmation Evaluation

On the locked 22-identity confirmation cohort (1,804 conditions, $N_{\text{timed}} = 1,711$ detected conditions), LS-Face and Direct SFace made identical final decisions on all 1,804 conditions (0 / 1,804 discordant decisions, 100.00% decision equivalence). Both

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systems correctly recognized 1,594 / 1,804 conditions (88.36%), with 117 rejections (6.49%) and 93 strict detector failures (5.16%) retained as failures (Table 7).

Table 7. Performance and latency comparison of Direct SFace and LS-Face on 1,804 locked confirmation conditions.

System	Accuracy	Mean latency	p50	p95	p99	Dual inf.	Early exits
Direct SFace	88.36%	8.300 ms	8.206 ms	9.183 ms	10.073 ms	0.00%	N/A
LS-Face	88.36%	7.015 ms	8.265 ms	11.704 ms	12.388 ms	18.06%	50.50%

Table 8. Decision complementarity matrix across 1,804 locked confirmation conditions.

SFace outcome	LBPH correct	LBPH incorrect	Total
SFace correct	1,156 (64.08%)	438 (24.28%)	1,594 (88.36%)
SFace incorrect	0 (0.00%)	210 (11.64%)	210 (11.64%)
Total	1,156 (64.08%)	648 (35.92%)	1,804 (100.0%)

Across the 1,711 timed conditions, LS-Face reduced mean recognition-stage latency from 8.300 ms to 7.015 ms, a 15.48% reduction (mean paired reduction 1.285 ms, identity-cluster bootstrap 95% CI: [1.088, 1.482] ms; paired difference -1.285 ms, 95% CI: [-1.482, -1.088] ms). On 50.50% of detected probes (864 / 1,711), LBPH terminated early at ~3.06 ms. Dual inference occurred on only 18.06% of probes (309 / 1,711) where acceptable-quality crops produced uncertain LBPH distances.

Table 8 reports the 2x2 decision complementarity matrix across all 1,804 locked confirmation conditions (1,156 both correct, 0 LBPH-only correct, 438 SFace-only correct, 210 both incorrect). Because LBPH-only correct is exactly 0, $P(\text{SFace correct} \mid \text{LBPH correct}) = 100.0\%$ within the evaluated cohort. This confirms that LBPH acts as a selective computational shortcut rather than an accuracy complement.

Workload Severity and Latency Profile. The operational benefit of LS-Face varies predictably across transformation severity tiers: Clean conditions achieved ~3.45 ms mean latency (58.4% reduction relative to Direct SFace), Light degradations averaged 6.176 ms (25.6% reduction), Medium degradations averaged 7.550 ms (9.0% reduction), and Heavy degradations averaged 7.722 ms (7.0% reduction). Standalone SFace maintains a tight single-path distribution (p50: 8.206 ms, p95: 9.183 ms, p99: 10.073 ms). LS-Face exhibits a wider tail on ambiguous acceptable-quality inputs (p50: 8.265 ms, p95: 11.704 ms, p99: 12.388 ms) due to the 18.06% dual-inference rate, in exchange for a 15.48% reduction in overall mean processing time.

5 Discussion

5.1 Quality-First Selective Computation

The experimental findings demonstrate that traditional sequential cascades are structurally inefficient because they execute classical descriptors indiscriminately on degraded frames that inevitably require deep-model fallback. Quality-first early bypass eliminates this bottleneck by routing degraded frames directly to SFace, cutting dual inference to 0.26% on LFW.

5.2 LBPH as a Computational Shortcut

Within the 1,804 locked confirmation conditions, every sample correctly identified by LBPH was also correctly identified by SFace (1,156 / 1,156), while SFace successfully resolved an additional 438 conditions where LBPH failed. This 100.0% subsumption demonstrates that classical descriptors do not expand the accuracy ceiling of modern deep face recognition. Rather, LBPH serves as a selective computational shortcut that resolves clear frontal inputs in ~ 3.06 ms instead of the 8.30 ms required for full deep neural network inference.

5.3 Robustness and Latency Trade-off

LS-Face matched standalone SFace transformation retention at 89.55% across all 41 LFW modifications (96.54% on 37 non-rotational conditions) and achieved a 15.48% mean latency reduction across 1,804 locked confirmation conditions. The primary operational trade-off is tail latency: while Direct SFace has a bounded single-path execution profile (p99: 10.073 ms), LS-Face incurs a heavier tail (p99: 12.388 ms) on ambiguous acceptable-quality frames that execute both recognizers. For edge access control, reducing average processing time from 8.300 ms to 7.015 ms reduces mean CPU computational load while keeping tail latency within interactive response bounds (< 15 ms).

5.4 Limitations

Several methodological boundaries constrain the interpretation of these results. First, the LFW robustness experiment measures controlled self-match retention under known synthetic transformations of enrolled images; it does not evaluate unconstrained cross-photo identification or false positive identification rates (FPIR). Second, the locked confirmation cohort comprises 22 local identities under synthetic transformations, and observations derived from the same source image remain correlated. Third, recorded latencies reflect recognition-stage CPU execution time (excluding detection and I/O) on an x86-64 processor; target embedded platforms (such as ARM-based microcontrollers or Raspberry Pi devices) may exhibit different relative compute ratios. Fourth, quality-check thresholds and the 0.05 margin threshold represent empirical heuristics

calibrated on development data. Finally, synthetic image transformations do not fully capture the complex multi-session lighting and environmental variations encountered in live production deployments.

6 Conclusion

6.1 Principal Findings

LS-Face demonstrates that quality-first selective computation enables hybrid face recognition cascades to achieve meaningful mean latency savings over direct deep neural network inference while preserving complete decision parity. By combining pre-descriptor quality screening with a compact LBPH fast path (36 KiB) and SFace fallback, LS-Face reduced mean recognition latency by 15.48% (7.015 ms vs. 8.300 ms; paired reduction 1.285 ms, bootstrap 95% CI: [1.088, 1.482] ms) across 1,804 locked confirmation conditions with 0 / 1,804 discordant decisions. On controlled LFW self-match transformations, LS-Face matched standalone SFace at 89.55% retention while restricting dual inference to 0.26%.

6.2 Architectural Implications

These findings establish that classical biometric algorithms remain valuable in the deep learning era not as accuracy complements, but as selective computational shortcuts. By formalizing early quality bypass and reserving deep inference for complex inputs, LS-Face provides a principled design framework for power-aware, low-latency biometric systems at the edge.

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